











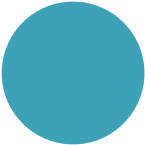
Reconstruct alignment using
Optimal Transport

Schiebinger et al. (2019)

Tong et al. (2020)

Bunne et al. (2021)

$$\inf_{T: T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x - T(x)\|_2^2 d\mu(x)$$



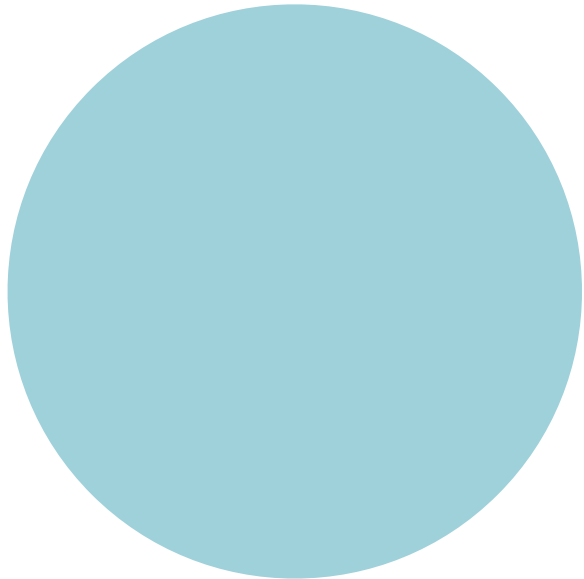




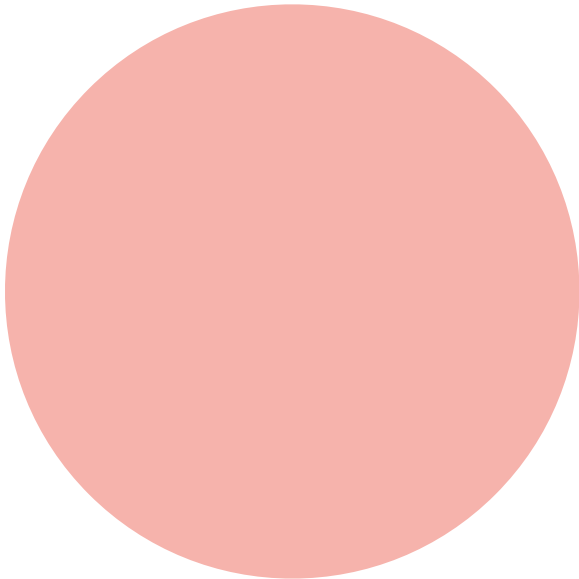


α

source



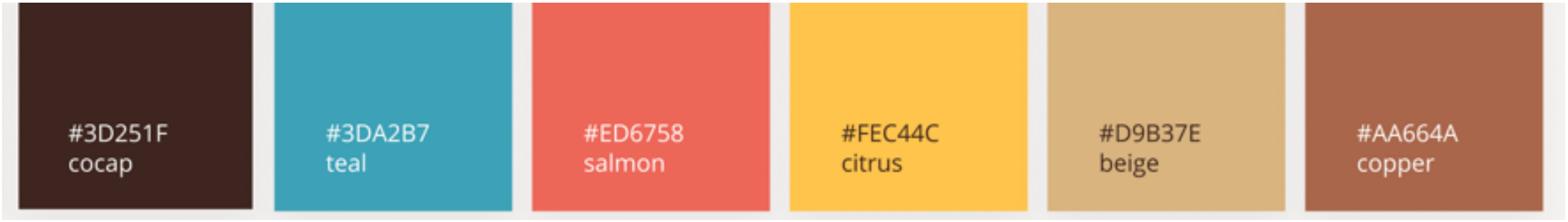
control



β

target

treated



#3D251F
cocap

#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper



Classic OT cannot account for this

























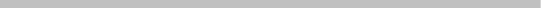










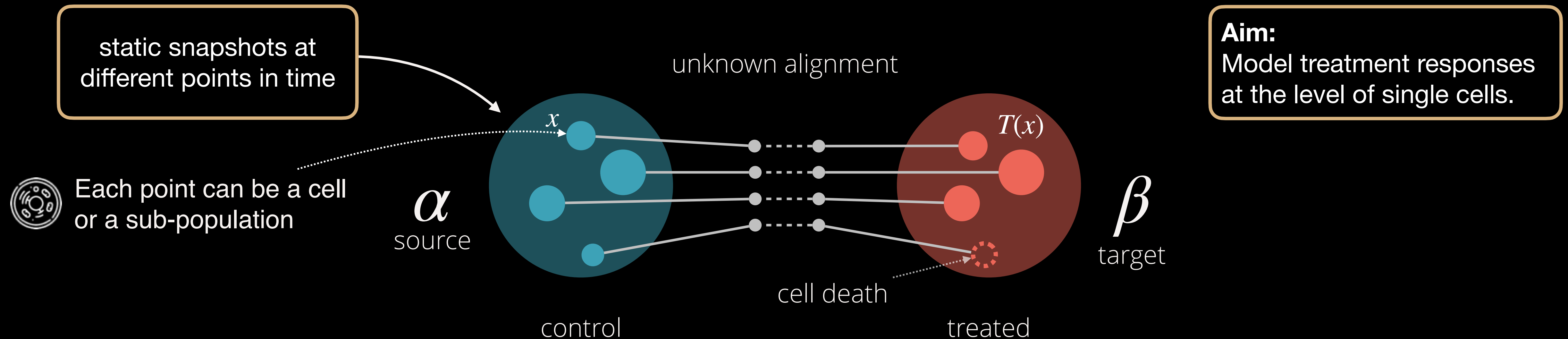






Unbalanced OT: Motivation

Modeling cell trajectories under various treatments. Challenge: Single-cell measurements typically destroy the cells.



Reconstruct alignment using
Optimal Transport
Schiebinger et al. (2019)
Tong et al. (2020)
Bunne et al. (2021)

$$\inf_{T: T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x - T(x)\|_2^2 d\mu(x)$$

Cells can die and proliferate as a response to treatment
 $\implies$ cell count (“mass”) might be different in α, β

Classic OT cannot account for this

Unbalanced OT: Motivation